Exercises

Exercise 8.1 · Prime or not?. Are the following integers prime? 211; 221; 1,001; 257.

Solution. 211 is prime (course). $221 = 13 \times 17$. $1{,}001 = 7 \times 11 \times 13$. For 257: none of 2, 3, 5, 7, 11, 13 divides it and $17^2 = 289 > 257$: it is prime.

Exercise 8.2 · Factorisation and divisors. Factorise 720 into primes. How many positive divisors does it have? Are there many divisors of 720 that are perfect squares?

Solution. $720 = 2^4 \times 3^2 \times 5$. Number of divisors: $(4+1)(2+1)(1+1) = 30$. A square divisor has even exponents: $2^{\beta_1}3^{\beta_2}5^{\beta_3}$ with $\beta_1 \in \{0,2,4\}$, $\beta_2 \in \{0,2\}$, $\beta_3 = 0$; there are 6 of them: 1, 4, 16, 9, 36, 144.

Beyond the syllabus · Least common multiple. The least positive common multiple of two nonzero integers, written lcm, is not part of the syllabus. It is used in the next exercise, which is not assessed.

Exercise 8.3 · GCD and LCM by factorisation. Factorise 1,260 and 1,050, then deduce their gcd and their least common multiple. Check that the product of the gcd and the lcm equals the product of the two numbers.

Solution. $1{,}260 = 2^2 \times 3^2 \times 5 \times 7$ and $1{,}050 = 2 \times 3 \times 5^2 \times 7$. GCD: $2 \times 3 \times 5 \times 7 = 210$; LCM: $2^2 \times 3^2 \times 5^2 \times 7 = 6{,}300$. And $210 \times 6{,}300 = 1{,}323{,}000 = 1{,}260 \times 1{,}050$.

Exercise 8.4 · Fermat and remainders.

1. Determine the remainder of 2^{2026} in the division by 13.
2. Prove that for every integer n, $n^{13} - n$ is divisible by 13.

Solution.

1. 13 is prime and does not divide 2: $2^{12} \equiv 1 \pmod{13}$. $2{,}026 = 12 \times 168 + 10$, so $2^{2026} \equiv 2^{10} = 1{,}024 \pmod{13}$. Now $1{,}024 = 13 \times 78 + 10$: the remainder is 10.
2. This is the first form of Fermat's little theorem, $n^{13} \equiv n \pmod{13}$.

Exercise 8.5 · Divisible by 42. Prove that for every integer n, $n^7 - n$ is divisible by 42.

Solution. $42 = 2 \times 3 \times 7$. By Fermat, $n^7 \equiv n \pmod{7}$. Modulo 2: $n^2 \equiv n$, so $n^7 \equiv n$. Modulo 3: $n^3 \equiv n$, so $n^7 = n^3 \times n^3 \times n \equiv n \times n \times n = n^3 \equiv n$. Thus 2, 3 and 7 divide $n^7 - n$; they are pairwise coprime, so (corollary of Gauss's theorem, applied twice) $42 \mid n^7 - n$.

Exercise 8.6 · Square of a prime number. Let p be a prime number greater than or equal to 5. Prove that 24 divides $p^2 - 1$.

Solution. $p^2 - 1 = (p-1)(p+1)$. Since p is odd, $p-1$ and $p+1$ are two consecutive even integers: one of them is a multiple of 4, so $8 \mid p^2 - 1$. Since p is not a multiple of 3, $p \equiv 1$ or $2 \pmod 3$, so $3 \mid p-1$ or $3 \mid p+1$. 8 and 3 are coprime: $24 \mid p^2 - 1$.

Exercise 8.7 · Mersenne numbers.

1. Using the factorisation of $z^m - 1$ (chapter 4), prove that if $n = ab$ with $a \geq 2$ and $b \geq 2$, then $2^a - 1$ divides $2^n - 1$.
2. Deduce that if $2^n - 1$ is prime, then n is prime.
3. Is the converse true? Compute $2^{11} - 1$ and check that 23 divides it.

Solution.

1. With $z = 2^a$: $2^n - 1 = z^b - 1 = (z-1)\left(z^{b-1} + \cdots + z + 1\right)$, so $2^a - 1 \mid 2^n - 1$.
2. If $n \geq 2$ is not prime, $n = ab$ as in 1, and $1 < 2^a - 1 < 2^n - 1$ is a divisor of $2^n - 1$, which is therefore not prime. Finally $2^1 - 1 = 1$ is not prime. By contraposition, if $2^n - 1$ is prime, n is prime.
3. No: 11 is prime, but $2^{11} - 1 = 2{,}047 = 23 \times 89$.

Problem

Problem · A miniature RSA cipher. Choose two primes $p = 5$ and $q = 11$, and let $n = pq = 55$ and $\varphi = (p-1)(q-1) = 40$.

1. Choose $e = 3$. Explain why e and φ are coprime, then find an integer $d \in \{1, \dots, 39\}$ such that $ed \equiv 1 \pmod{40}$.
2. Encryption: a message is an integer $m \in \{0, \dots, 54\}$; it is encrypted as the remainder c of m^e modulo n. Encrypt $m = 8$.
3. Decryption: we take for granted for now that $c^d \equiv m \pmod{n}$. Check with a calculator or Python that $17^{27} \equiv 8 \pmod{55}$.
4. We prove that $m^{ed} \equiv m \pmod{55}$ for every integer m.
   (a) Write $ed = 1 + 40k$ with k a natural number.
   (b) If $5 \nmid m$, prove that $m^{40k} \equiv 1 \pmod{5}$; deduce $m^{ed} \equiv m \pmod{5}$. What happens if $5 \mid m$?
   (c) Establish likewise $m^{ed} \equiv m \pmod{11}$, then conclude.
5. Why does the security of this cipher rest on the difficulty of factorising n when p and q are very large?

Solution · Problem.

1. $40 = 2^3 \times 5$ and 3 does not divide 40: they are coprime. $3 \times 27 = 81 = 2 \times 40 + 1$: $d = 27$.
2. $8^3 = 512 = 9 \times 55 + 17$: $c = 17$.
3. We do get 8 (for example `pow(17, 27, 55)`).
4. (a) $ed = 81 = 1 + 40 \times 2$: $k = 2$.
   (b) By Fermat, $m^4 \equiv 1 \pmod{5}$, so $m^{40k} = \left(m^4\right)^{10k} \equiv 1$ and $m^{ed} = m \times m^{40k} \equiv m \pmod{5}$. If $5 \mid m$, both sides are congruent to 0.
   (c) Modulo 11: if $11 \nmid m$, $m^{10} \equiv 1$, so $m^{40k} = \left(m^{10}\right)^{4k} \equiv 1$; if $11 \mid m$, both sides are zero modulo 11. Thus 5 and 11 divide $m^{ed} - m$ and are coprime: $55 \mid m^{ed} - m$.
5. Computing d by this method requires $\varphi = (p-1)(q-1)$, hence p and q. Knowing only n and e, factorising n would be enough to find d. For well-chosen primes with several hundred digits, no known method factorises n in a reasonable time on current computers. The security of RSA rests on this assumed difficulty: it has not been proved that every way of decrypting goes through factorising n.

Teacher note · Status of the problem. RSA is a "possible problem" of the syllabus. Have the encryption and decryption functions written in Python and tested on the 55 possible messages. Question 5 gives an idea of security; it does not call for a proof.